

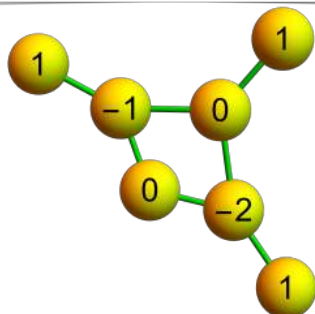
Poincare-Hopf

$$\sum_{v \in V} i_g(v) = \chi(M)$$

$$S_g^-(v) = \{w \in S(v), g(w) < g(v)\}$$

$$i_g(v) = 1 - \chi(S_g^-(v))$$

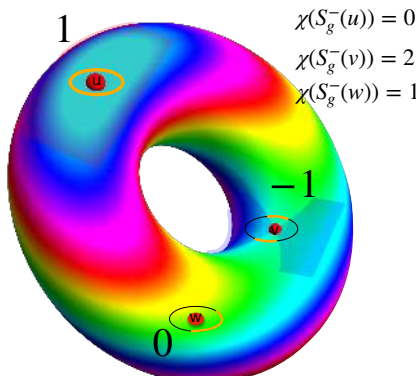
index



Classical

$$\sum_{v \in V} i_g(v) = \chi(M)$$

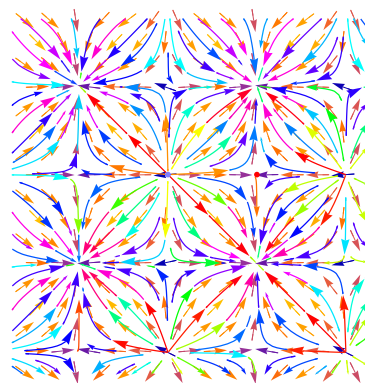
$$i_g(v) = 1 - \chi(S_g^-(v))$$



Morse

Classical Morse functions:
 $i_g(v) = (-1)^m$ for Morse index m .

Discrete Morse functions: $S_g^-(v)$
 is a $(k-1)$ -sphere: then $m=k-1$

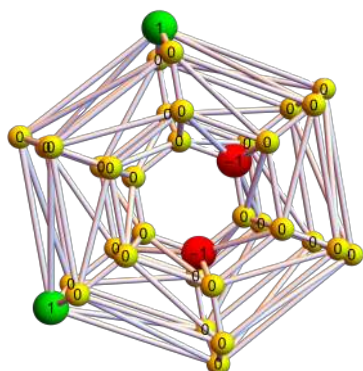


$i=-1$ for saddle points
 $i=1$ for maxima or minima

Example

$$\sum_{v \in V} i_g(v) = \chi(G)$$

$$i_g(v) = 1 - \chi(S_g^-(v))$$

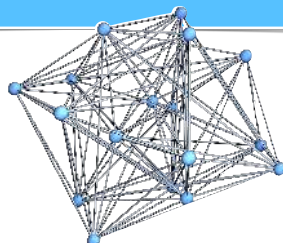


Functional

$$f_G(t) = 1 + t \sum_{v \in V} f_{S_g^-(v)}(t)$$

$$i_g(v) = f_{S_g^-(v)}(-1)$$

```
S[G__g__v_] := Module[{H = VertexDelete[NeighborhoodGraph[G, v, v],
Subgraph[H, Select[VertexList[H], ((# /. g) < (v /. g)) &]]];
f[G__g__t_] := Module[{V = VertexList[G, v], Clear[t];
1 + t*Sum[v = V[[k]]; f[S[G, g, v], g, t], {k, Length[V]}];
G = GraphJoin[CompleteGraph[{2, 3, 5}], CycleGraph[7]];
V = VertexList[G]; g = Table[V[[k]] -> Random[], {k, Length[V]}];
Print["Euler = ", 1 - f[G, g, t]/t -> 1]; Print[Simplify[f[G, g, t]]];
GraphPlot3D[G]
```

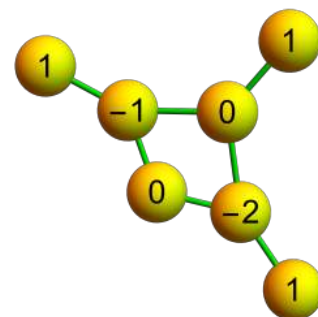


$$f_G(t) = 1 + 7t + 7t^2$$

$$f_G(t) = 1 + t \sum_{v \in V} f_{S_g^-(v)}(t)$$

$$i_g(v) = f_{S_g^-(v)}(-1)$$

$$\begin{bmatrix} 3t + 1 \\ t + 1 \\ 2t + 1 \\ t + 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$



Vector Field

$$F : G \rightarrow V \text{ such that } F(x) \in x$$

This and a map $F : V \rightarrow G$
 such that $v \in F(v)$ defines
 a permutation in G , a flow.

$$i_F(v) = 1 - \chi(S_F^-(v))$$

where $S_F^-(v) = F^{-1}(v)$

$$\sum_{v \in V} i_F(v) = \chi(M)$$

Proof: $F(x)$ plays the role of
 maximim, if a function g is
 given.

Digraph

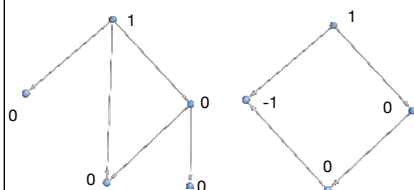
A digraph is locally irrotational, if
 $x \rightarrow y$ defines a total order each
 complete subgraph.

$$i(v) = 1 - \chi(S^-(v))$$

For a locally irrotational digraph:

$$\sum_{v \in V} i(v) = \chi(M)$$

Proof: The vertices in every maximal
 simplex are totally ordered. Assign
 $F(x)=v$, the maximum of this order.



Classical

Poincare (1895)

in two dimensions

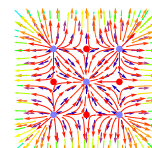
Hopf (1926)

in general

Poincare



Hopf



Discrete

Glass (1973)

graphs embedded in
 2-manifolds

Knill (2012):

arbitrary dimensions

Knill (2019):

functional version

Knill (2019):

for vector fields
 or directed graphs